

RPM 2.0: RF-based Pose Machines for Multi-Person 3D Pose Estimation

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Abstract—Advanced human sensing technologies based on radio frequency (RF) signals have gained widespread attention in recent years. However, due to the sparsity and incompleteness of RF signals, fine-grained RF-based multi-person 3D pose estimation has progressed more slowly. In this paper, we present RF-based Pose Machine (RPM 2.0) for multi-person 3D pose estimation using RF signals. Specifically, we first develop a lightweight anchor-free detector module to locate and crop regions of interest from horizontal and vertical RF signals. Afterward, we treat the horizontal and vertical millimeter-wave radars as “RF cameras” with different viewing angles and propose a Multi-view Fusion Network to unproject the RF signals into a unified latent feature space, and then calculate the correlation for weighted fusion. Finally, a Spatio-Temporal Attention Network is designed to reconstruct the multi-person 3D skeleton sequences, in which the spatial attention module is proposed to recover invisible body parts using non-local correlations among joints and the temporal attention module refines the 3D pose sequences using temporal coherency learned from frame queries. We evaluate the performance of the proposed RPM 2.0 and state-of-the-art methods on a large-scale dataset with multi-person 3D pose labels and corresponding radar signals. The experimental results show that RPM 2.0 outperforms all of the baseline methods, which locates multi-person 3D key points with an average error of 73mm and generalizes well in new data such as occlusion, low illumination.

Index Terms—RF sensing, 3D Human Pose Estimation, Deep Learning, Smart Homes.

I. INTRODUCTION

MULTI-PERSON 3D pose estimation, which aims to locate the 3D joints of each person in a scene, is a fundamental task that has a great impact on various application scenarios such as mixed reality, enterprise security, smart homes, and medical care. With the emergence of deep learning,

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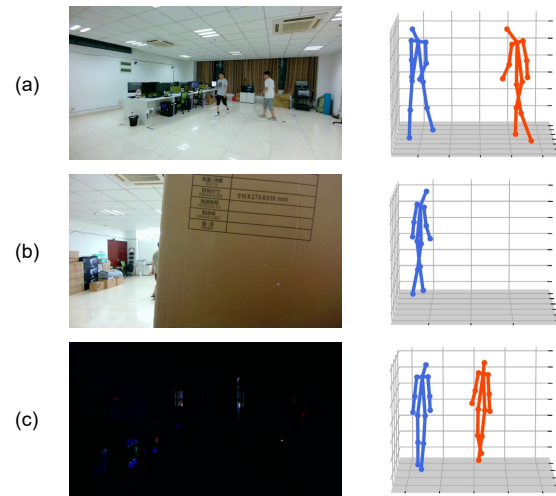


Fig. 1. Our proposed RPM 2.0 can reconstruct accurate multi-person 3D skeletons under different environments: (a) basic environment, (b) occlusion, (c) bad illumination.

multi-person 3D pose estimation based on visual images or videos has made tremendous progress. However, camera-based multi-person 3D pose estimation methods suffer from severe performance degradation in the face of complex scenes, such as occlusion, bad illumination, blurry, etc. Moreover, the extensive use of cameras can bring privacy concerns, which limit the scope of application.

Meanwhile, the past decade has witnessed significant progress in using different sensing modalities to expand beyond the capabilities of cameras. Specifically, RF signals, due to the ability to traverse occlusions and be unaffected by light, show great application value. Previous works [1]–[4] demonstrate that RF signals carry an impressive amount of information that can be used to enable various human sensing tasks such as breathing and heartbeats inference, gesture recognition, indoor localization, and tracking. These methods are able to cover a large space, even with occlusion, and are privacy-preserving because they do not record visual data. However, for some fine-grained tasks, such as 3D human pose estimation, it is still challenging to achieve high accuracy due to the much lower spatial resolution of RF signals.

The main challenge in inferring 3D skeletons solely from RF signals is the specularly of the human body with respect to RF signals. When exposed to radar range, the target behaves like a reflector other than a scatterer. In particular, a single RF

snapshot can only capture a subset of the limbs and misses body parts whose orientation deflects the signal away from the radar. To further complicate this issue, there is no direct relationship between the RF signal reflected by the body and the key points, which means that we cannot know which body part is reflecting the signal back. The fact that a single RF snapshot only contains limited information about an unknown subset of human body parts makes the RF-based 3D pose estimation a highly unconstrained problem.

In this paper, we introduce RPM 2.0, a novel framework that infers accurate multi-person 3D skeletons from commercial millimeter-wave radars. Following the divide-and-conquer paradigm, we first detect multiple people over time in 3D space and then estimate the corresponding 3D skeletons. Considering the RF signal is sparse in which only a few objects have strong responses, RPM 2.0 develops an anchor-free human detector to estimate the bounding box centers and corresponding box sizes directly, which is more simpler and efficient. To obtain more robust feature representations, RPM 2.0 treats the two millimeter wave radars as RF cameras with different viewpoints and uses a Multi-view Fusion Network to fuse the features in a unified latent space based on the inter-radar transformation relationship. Afterward, RPM 2.0 zooms into each detected trajectory which contains multiple RF snapshots over time to estimate 3D skeletons. In order to overcome the specularity problem, we proposed a Spatio-Temporal Attention Network to extract the maximum useful information from both spatial and temporal dimensions. Specifically, the spatial attention module is designed to recover invisible body parts using non-local correlations among joints, while the temporal attention module refines the 3D pose sequences using temporal coherency learned from frame queries.

To train our RPM 2.0 model, we conduct a multi-modality data collection system by designing a multi-camera system to generate 3D skeletons labels of multiple people and a dual-radar system to maintain synchronized RF signals. We train and test RPM 2.0 on the collected large-scale dataset. The evaluation results show that RPM 2.0 is able to estimate multi-person 3D poses with an average error of $73mm$, which outperforms the state-of-the-art methods. As shown in Fig. 1, RPM 2.0 maintains its performance when generalized to new data such as occlusion, low illumination. We conduct various ablation studies to demonstrate the effectiveness of individual components in RPM 2.0.

II. RELATED WORK

A. Camera-based 3D Human Pose Estimation

Fundamentally, camera-based human pose estimation (HPE) aims to reconstruct the 3D skeleton from 2D data. According to different input data, camera-based 3D pose estimation can be divided into two categories: monocular 3D HPE methods and multi-view 3D HPE methods.

With the rapid growth of deep learning, monocular 3D HPE approaches [5]–[8] have achieved impressive performance. Some methods [9]–[11] force models to map 2D pixels to 3D poses directly. Albiero et al. [12] proposed a method for 3D face pose estimation without face detection or landmark

localization, in which Faster R-CNN-based architecture was applied to directly extract 6DoF, 3D face pose estimation, without requiring face detection. Cao et al. [13] adopted directional statistics in the task of pose estimation and proposed a ASG-based Label Distribution Learning method for estimating facial pose which addressed the issue of biased expectation. However, recovering 3D poses from one 2D image is an ill-posed task because a single image may correspond to many possible 3D poses, which makes these networks difficult to learn. Thus, various prior constraints are utilized to reduce the uncertainty. Fang et al. [14] and Xu et al. [15] incorporated kinematics constraints in 3D pose estimation. Other works [16]–[18] applied Graph Convolutional Networks (GCN) to encode the relationships between joints. These efforts tried to fully exploit the potential information in a single image, which is however not sufficient for solving the ill-posed problem. Many works [19]–[21] then focused on utilizing temporal consistency in videos. Pavllo et al. [19] proposed a temporal CNN model to estimate 3D poses from 2D pose sequences. Motivated by [19], Chen et al. [21] added a bone-length module and a bone-direction module to better model the temporal coherency. Zheng et al. [20] applied a spatio-temporal transformer to comprehensively model the human joint relations within each frame as well as the temporal correlations across frames from the input 2D pose sequences. In summary, monocular 3D pose estimation is easier to implement because it does not require multiple camera parameters obtained by calibration, but these methods do not deal well with occluded scenes and have depth ambiguity.

Multi-view 3D Human Pose Estimation, on the other hand, can better tackle these issues. Recent works on single-person scenes utilized either intrinsic and extrinsic camera parameters for feature fusion [22], [23] and triangulation [24], [25], or pictorial structure models for 3D pose reconstruction [22]. Nevertheless, reconstructing multi-person 3D poses using multi-view data remains challenging. Volumetric approaches like VoxelPose [26] formed a comprehensive feature volume through heatmap un-projection and then localized each person instance for 3D pose estimation. Other research works [27]–[30] aggregated 2D skeletons estimated by existing 2D HPE methods [31]–[33] and reconstructed 3D poses by triangulation or 3D pictorial structure.

Monocular and multi-view camera-based methods exploit various clues to achieve 3D HPE tasks in different scenes. However, due to the poor penetration of visible light, camera-based HPE methods are not good at inferring the 3D human pose in scenes such as occlusion and low illumination. Our proposed RPM 2.0, on the other hand, can meet these challenges thanks to the use of more penetrating RF signals.

B. Wireless Sensing

Significant research efforts [3], [34]–[48] have been spent on enhancing the perceptual capabilities of the model using different sensing modalities. Among them, RF signals have gained more attention due to their better penetration and resolution. Many works utilized RF signals to enable various human sensing tasks such as physiological sensing [2], [3],

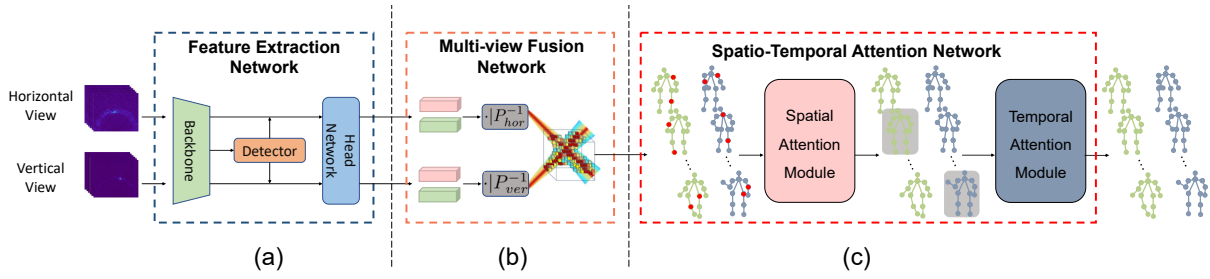


Fig. 2. The proposed RPM 2.0 framework for Multi-Person 3D pose estimation. (a) Horizontal and vertical RF signals are fed to the Feature Extraction Network, in which features are cropped based on the bounding box results produced by the human detector. (b) Then the Multi-view Fusion Network transforms features from horizontal and vertical radars into a unified latent space and obtains aggregated features. (c) Fused features are randomly masked (red dots) along the joint axis and fed to the Spatial Attention Module models construct body parts that cannot be captured by RF snapshots based on the non-local skeleton relationships. Finally the Temporal Attention Module refines multi-person 3D poses based on temporal coherency learned from partially masked (gray skeletons) frame queries.

[42], gesture recognition [1], human detection and tracking [4], [49] and etc. However, due to the scarcity of large-scale labeled data and the difficulty of fine-grained human body perception, only a few works [50]–[53] have investigated RF-based 3D HPE task. RFPose3D [50] first used a region proposal network (RPN) to determine the location of the human body, and then classified each voxel in the space based on the extracted features to discriminate whether it is a key point of the human body. Sengupta et al. [51] proposed a simple network, mm-Pose, to reconstruct the 3D pose of a single person from point cloud data. Yu et al. [53] proposed a lightweight model, RFPose-OT, to enable RF-based 3D HPE from the horizontal RF signals using optimal transport theory. Xie et al. [52] proposed RPM, which introduced spatio-temporal attention to significantly improve the accuracy of single-person 3D pose estimation.

After witnessing the excellent performance of RPM [52] in single-person 3D pose estimation based on RF signals, we propose RPM 2.0 for accurate multi-person 3D pose estimation. Although both RPM 2.0 and the previous RPM [52] used spatio-temporal attention to predict 3D human poses, RPM 2.0 made many improvements for multi-person pose estimation using RF signals. Specifically, in this work, we treat the horizontal and vertical mmwave radars as “RF cameras” with different viewing angles. We first propose a lightweight anchor-free detector module to locate and crop regions of interest from RF signals. Afterward, a Multi-view Fusion Network is proposed to unproject the RF signals into a unified latent feature space, and then calculate the correlation for weighted fusion. Finally, a Spatio-Temporal Attention Network is designed to reconstruct the multi-person 3D skeleton sequences, in which the spatial attention module is proposed to recover invisible body parts using non-local correlations among joints and the temporal attention module refines the 3D pose sequences using temporal coherency learned from frame queries. In order to extend the Spatio-Temporal Attention Network to the task of multi-person pose estimation, we further propose a bipartite-matching-based training strategy to improve the training efficiency of RPM 2.0.

C. Attention in Computer Vision

In the field of computer vision, attention mechanisms have demonstrated effectiveness and have been widely utilized across various downstream tasks. Zhu et al. [54] proposed an end-to-end flow correlation tracking framework in which spatial-temporal attention was utilized to aggregate frames adaptively for consequent correlation filter tracking. Yan et al. [55] proposed STAT for video captioning, which considered both the spatial and temporal structures in a video to automatically select the significant regions in the most relevant temporal segments for word prediction. Fu et al. [56] proposed the SPatial-Temporal Attention module to address the large-scale video-based person Re-ID problem, in which the 2-D spatial-temporal map is used to calculate the more robust clip-level feature representation without using any additional parameters. Yan et al. [57] introduced a hierarchical attention fusion network for geo-localization. The multi-scale feature maps is extracted for hierarchical attention fusion to produce robust image representation for landmarks with different scales. Cui et al. [58] proposed a framework called TF-Blender for video object detection in which a temporal relation module is designed to manage temporal information and a feature adjustment module to add constraints in feature learning to preserve spatial information. Liu et al. [59] proposed Denser-Net for visual localization, in which feature attention filter implicitly encodes spatial concept into the parameters of the attention features to represent the irregular landmarks flexibly.

Attention mechanisms have been widely applied to various tasks in the field of computer vision. However, in the field of wireless sensing, exploration of attention mechanisms is still limited. In this paper, we explore spatio-temporal attention mechanisms and introduce them into RF-based multi-person 3D pose estimation task. Our proposed RPM 2.0 includes a Spatio-Temporal Attention Network to extract the maximum useful information from both spatial and temporal dimensions. Specifically, the spatial attention module is designed to recover invisible body parts using non-local correlations among joints, while the temporal attention module refines the 3D pose sequences using temporal coherency learned from frame queries. Without bells and whistles, RPM 2.0 achieves state-of-the-art performance in multi-person 3D HPE tasks.

III. PRIMER

A. Multi-antenna FMCW Radar

Frequency Modulated Continuous Wave (FMCW) radar has been widely used for signal transceiving. The radar is equipped with an antenna array to separate the signal from different directions. By combing FMCW and antenna array, the RF signals from the location (x, y, z) can be extracted as

$$r(x, y, z, t) = \sum_{k=1}^K \sum_{m=1}^M s_{k,m,t} \cdot e^{j2\pi \frac{d_m(x,y,z)}{\lambda_k}} \quad (1)$$

where $s_{k,m,t}$ denotes the k -th sample of FMCW sweep on the m -th antenna at time t , λ_k is the signal wavelength, $d_m(x, y, z)$ denotes the round-trip distance from (x, y, z) to radar. As demonstrated in [50], the complexity of processing 4D tensor is extremely high, which can not be handled by major machine learning platforms. Therefore, similar to [50], we decompose the signal into horizontal and vertical planes (e.g., Fig. 3). The input data for neural network then takes the form of two-dimensional heatmaps, which can be given by

$$r_{hor}(x, y, t) = \sum_{k=1}^K \sum_{m=1}^M s_{k,m,t} \cdot e^{j2\pi \frac{d_m(x,y)}{\lambda_k}}, \quad (2)$$

$$r_{ver}(y, z, t) = \sum_{k=1}^K \sum_{m=1}^M s_{k,m,t} \cdot e^{j2\pi \frac{d_m(y,z)}{\lambda_k}}. \quad (3)$$

To address the multi-path interference in indoor environment [4], we subtract consecutive signal measurements to suppress static multi-path while retain human reflections.

B. Multi-camera System

Since motion capture system like VICON [60] is not portable and cannot capture person without markers, we build a simple and effective multi-camera system using Raspberry Pi's to generate 3D skeleton labels. As shown in Fig. 3, our multi-camera system consists 13 Raspberry Pi 4 with wide-angle camera lenses.

Once deployed, the multi-camera system simultaneously captures people from different viewpoints and generates 2D

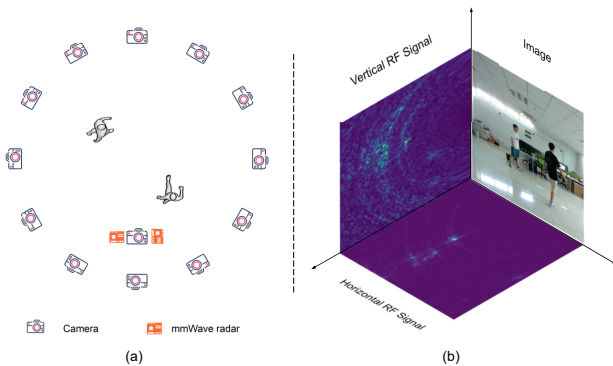


Fig. 3. The data collection system. (a) 13 cameras are utilized to generate 3D skeleton labels and 2 mmWave radars are used to obtain RF signals in horizontal and vertical planes. (b) The captured RGB image and corresponding RF heatmaps at the same timestamp.

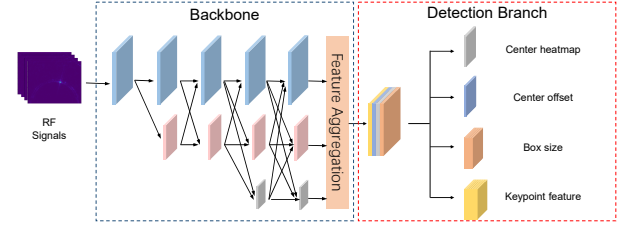


Fig. 4. The proposed Feature Extraction Network. The multi-scale modeling is applied to avoid the loss of information when processing small objects. The anchor-free detection branch contains an anchor-free detector which represents the target as combination of a single point and corresponding bounding box and a keypoint head to generate feature maps for 3D pose estimation.

poses through Alphapose [31]. Using the 2D poses P_{2D} from multiple views and the projection matrices [61] of cameras, we can perform triangulation to get accurate 3D skeletons:

$$P_{3D} = \arg \min_{P_{3D}} \sum_{i \in I} \|T^i P_{3D} - P_{2D}^i\|_2^2 \quad (4)$$

where T^i is the projection matrix of camera i .

IV. METHOD

The design philosophy of our framework is to adaptively reconstruct 3D skeletons from an arbitrary number of persons and arbitrary RF sequence lengths using horizontal and vertical radars. As shown in Fig. 2, the overall pipeline is to detect person instances and crop the feature regions first, then obtain the fused features, and finally predict 3D skeletons based on spatio-temporal dependencies. The RPM 2.0 consists of Feature Extractor, Multi-view Fusion Network, and Spatio-Temporal Attention Network which are introduced as follows.

A. Feature Extractor

As shown in Fig. 4, the Feature Extractor consists of a multi-scale backbone for obtaining robust feature maps and a detection branch for locating and cropping person instances.

1) *Backbone*: In radar signals, the scale of the target can be extremely small. Therefore multi-scale modeling is crucial to avoid the loss of information when the network performs downsampling operations. We use a pre-trained HRNet-18 [62] as the backbone in the feature extractor. Denote the size of each input RF image as $H_{input} \times W_{input}$, then the output feature map size is $H \times W$, where $H = H_{input}/4$ and $W = W_{input}/4$.

2) *Detection Branch*: To reconstruct 3D skeletons of multiple people from the RF signal, we follow the divide-and-conquer paradigm by first locating and cropping people regions and then zooming into each region to estimate 3D joints for each individual. Previous work [50] uses a popular anchor-based module, region proposal network (RPN), to generate people regions. However, RPN first produces an extensive number of potential object bounding boxes and then removes duplicated detections by computing confidence scores and bounding box IoU, which is complex. Considering that the RF signal is usually sparse in which only a few objects

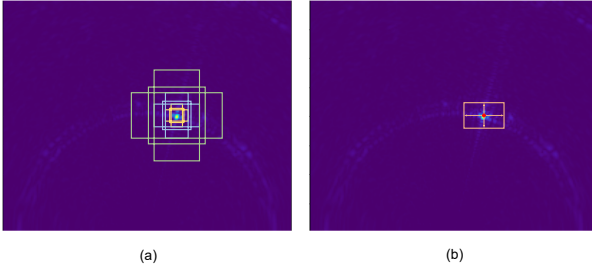


Fig. 5. Different detection methods. (a) Typical anchor-based detector which generates massive duplicated detections. (b) Center-based detector that models the target as a center point of its bounding box. Best viewed in color.

have strong responses, we provide a much simpler and more efficient alternative. As shown in Fig.5, we represent the target in the RF signal as a single point at its bounding box center. The object properties, such as box weight and height, and class scores are then predicted directly from the RF features at the center location. Specifically, several parallel heads are followed by the backbone to estimate the object center heatmap, center offset, bounding box size, and keypoint heatmap, respectively.

Center Heatmap Head. This head is designed to locate the center of each person instance. Since horizontal RF images are projections of RF signals on a plane parallel to the ground, which can provide the human body position, the center heatmap head only estimates the human centers with horizontal RF images for simplicity. The corresponding vertical locations can be determined with the horizontal positions and a fixed height range. Given the output feature map from the backbone, the center heatmap head produces a $1 \times H \times W$ heatmap to represent all the human centers.

For each RF image, we formulate the bounding box label as:

$$b_{GT}^i = (x_1^i, y_1^i, x_2^i, y_2^i), \quad i \in [1, 2, \dots, K] \quad (5)$$

where K is the number of person instances. We can compute the box center (c_x^i, c_y^i) as $c_x^i = (x_1^i + x_2^i)/2$ and $c_y^i = (y_1^i + y_2^i)/2$. Then its location on the down-sampled center heatmap is $(\hat{c}_x^i, \hat{c}_y^i) = (\lfloor c_x^i/4 \rfloor, \lfloor c_y^i/4 \rfloor)$. Afterward, the ground-truth heatmap response can be given by:

$$\hat{H}_{x,y} = \sum_{i=1}^K \exp\left(-\frac{(x - \hat{c}_x^i)^2 + (y - \hat{c}_y^i)^2}{\sigma^2}\right) \quad (6)$$

where σ represents the standard deviation. The loss function is defined as pixel-wise MSE loss:

$$\mathcal{L}_{center} = \frac{1}{XY} \sum_{xy} (\hat{H}_{xy} - H_{xy})^2 \quad (7)$$

where H is the estimated center heatmap.

Center Offset Head. Since the final feature map is downsampled by a factor of four, the estimation of the centers will have a quantization error of up to 4 pixels. Therefore, we introduce an offset head to ensure a more accurate prediction of the center position. For each person instance, the GT offset is $\hat{o}^i = (c_x^i/4 - \lfloor c_x^i/4 \rfloor, c_y^i/4 - \lfloor c_y^i/4 \rfloor)$. The dimension of the

output offset heatmap is $2 \times H \times W$. Denote the estimated offset at the corresponding location as $\hat{o}^i$, then the offset loss is computed as follows:

$$\mathcal{L}_{offset} = \frac{1}{K} \sum_i^K \|\hat{o}^i - o^i\|_1 \quad (8)$$

Box Size Head. The box size head is designed to estimate the size of the bounding box. To generate the bounding box more flexibly, we predict the offset of the upper left and lower right points relative to the center point rather than the width and height. Denote the output as $\hat{S} \in \mathbb{R}^{4 \times H \times W}$. The GT size is $\hat{s}^i = (\hat{c}_x^i - x_1^i, \hat{c}_y^i - y_1^i, x_2^i - \hat{c}_x^i, y_2^i - \hat{c}_y^i)$. The box size loss is then written as:

$$\mathcal{L}_{size} = \frac{1}{K} \sum_i^K \|\hat{s}^i - s^i\|_1 \quad (9)$$

where s^i is the estimated box size.

Keypoint Heatmap Head. The keypoint heatmap head aims to generate feature maps for the subsequent 3D pose estimation task. To achieve this goal, we apply a 3×3 convolution layer (with 256 channels) to the output feature of HRNet18 backbone, followed by a 1×1 convolution layer (with 128 channels) to generate the final feature map of each RF image.

B. Multi-view Fusion Network

Denote the resulting feature maps as $E_{hor} \in \mathbb{R}^{128 \times H \times W}$ and $E_{ver} \in \mathbb{R}^{128 \times H \times W}$. We first crop regions of each target with the bounding box results obtained from the detection branch. In particular, given the decoded detection bounding box $b_{pr}^i = (x_1^i, y_1^i, x_2^i, y_2^i)$, we apply RoIAlign to get the cropped region.

One critical problem is how to fuse the cropped features from horizontal and vertical RF images. Although the imaging principle of radar is different from that of a conventional camera, we can still analogize our task to getting multi-person 3D poses from the “RF cameras” with multiple views. In this work, we designed a Multi-view Fusion Network to map the cropped RF features from multiple radar views to a unified latent representation of 3D skeletons. Specifically, we project the cropped feature maps E_i to a common canonical representation by conditioning them on a projection matrix P_i^{-1} that maps RF image coordinates to the world coordinates:

$$E_i^w = \text{Proj}(E_i | P_i^{-1}) \quad (10)$$

Since the projection matrix P_i^{-1} of “RF camera” is hard to obtain by calibration like conventional cameras, we implement the projection process implicitly through a convolution layer. We encode the bounding box b^i , which implicitly contains information about the coordinate transformation, and combine it with the RF image as a feature embedding:

$$E_i^h = E_i + \text{Embed}(b^i) \quad (11)$$

where Embed is implemented by a linear layer. Afterward, we can map the feature embedding E_i^h to the same canonical representation via a shallow 1D convolutional neural network:

$$E_i^w = \text{Conv1D}(E_i^h) \quad (12)$$

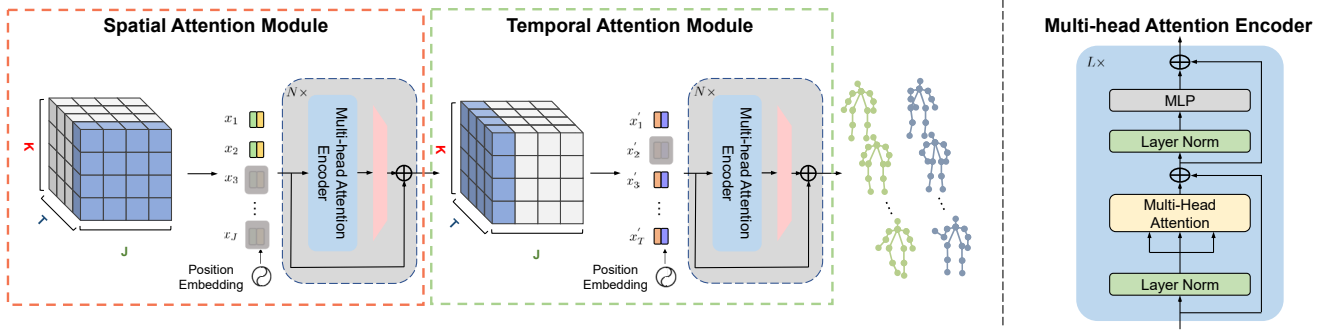


Fig. 6. The proposed Spatio-Temporal Attention Network. The fused feature from the Multi-view Fusion Network is first transformed into multiple joint queries x_i of each body joint i . Then the joint queries are randomly masked and fed to the spatial attention module to model non-local skeleton relationships. Afterward, the output feature is transformed into queries x'_t along the time axis of each frame t and randomly masked. The temporal attention module finally refines the 3D skeleton sequences based on temporal coherency. We apply a progressive dimensionality reduction scheme to both the spatial and temporal attention modules to gradually reduce the hidden dimensions so that our model can regress the 3D coordinates of the human skeletons directly.

Now we can aggregate the projected feature maps from all views to form an input to the subsequent modules that don't depend on the number of "RF camera" views. We study three different aggregation methods:

1) *Concatenation of the feature:*

$$\mathbf{E}^{input} = \text{Concat}(\mathbf{E}_{hor}^w, \mathbf{E}_{ver}^w) \quad (13)$$

2) *Raw summation of the feature:*

$$\mathbf{E}^{input} = \sum_v \mathbf{E}_v^w, \quad v \in [hor, ver] \quad (14)$$

3) *Summation of the feature with confidence:* Here we first compute the softmax for each pixel across the concatenated feature maps $\mathbf{E}^{concat} \in \mathbb{R}^{V \times C \times H \times W}$ as the confidence distribution:

$$\mathbf{W} = \exp(\mathbf{E}^{concat}) / \sum_v (\exp(\mathbf{E}_v^{concat})) \quad (15)$$

Next, we weight and sum the features transformed from different views:

$$\mathbf{E}^{input} = \sum_v (\mathbf{E}^{concat} \cdot \mathbf{W}) \quad (16)$$

C. Spatio-Temporal Attention Network

Like conventional cameras, radar has blind spots when capturing information about the human body. RF specularly is a physical phenomenon in which the object is like a reflector other than a scatter. Therefore, a single RF snapshot can only capture a subset of the limbs and misses body parts whose orientation deflects the signal away from the radar at that time. To reconstruct accurate 3D skeletons of multi-person from the sparse and incomplete RF signals, we propose a Spatio-Temporal Attention Network to model spatio-temporal relationships.

1) *Revisit Multi-head Self-attention:* Denote the embedded feature as $\mathbf{H}$. The scaled dot-product attention can be computed by mapping a query matrix $\mathbf{Q}$, key matrix $\mathbf{K}$ and value matrix $\mathbf{V}$ to an attention matrix, which can be written as:

$$\mathbf{Q} = \mathbf{H}\mathbf{W}_Q, \mathbf{K} = \mathbf{H}\mathbf{W}_K, \mathbf{V} = \mathbf{H}\mathbf{W}_V \quad (17a)$$

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{C}}\right) \mathbf{V} \quad (17b)$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N \times C}$, $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{C \times C}$. A scaling factor of $1/\sqrt{C}$ is utilized for normalization.

The Multi-head Self-attention layer model the attention with multiple heads which computes the scaled dot-product attention in Eqn. (17) in parallel:

$$\text{MHAttn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\mathbf{O}_1, \dots, \mathbf{O}_h) \mathbf{W}_{out} \quad (18a)$$

$$\mathbf{O}_i = \text{Attn}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i), \quad i \in [1, 2, \dots, h] \quad (18b)$$

where h is the number of attention heads and $\mathbf{W}_{out} \in \mathbb{R}^{(h \cdot C) \times C}$ is the projection matrix.

2) *Spatial Attention Module (SAM):* Denote the output feature of the Multi-view Fusion Network as $\mathbf{E}^{input} \in \mathbb{R}^{T \times M \times J \times C}$, where T is the RF sequence length, M is the maximum number of person instance in each RF image, J is the number of joints and C is the feature channel. Since major machine learning platforms (PyTorch, Tensorflow) cannot support convolution operation on such 4D data, we decompose the 4D tensor to lower dimensions for simplicity. In particular, when modeling the spatial non-local relationships of joints, we focus on the joint dimension and transform the original feature as follows:

$$\mathbf{E}^{input} \in \mathbb{R}^{T \times M \times J \times C} \Rightarrow \mathbf{X}_{spatial} \in \mathbb{R}^{D \times J \times C} \quad (19a)$$

$$\text{where } D = T \cdot M \quad (19b)$$

in which each feature $\mathbf{x}_{spatial} \in \mathbb{R}^{J \times C}$ of $\mathbf{X}_{spatial}$ is the joint queries of single person in each RF image.

Furthermore, we design a Spatial Mask to simulate the specularly of human body with respect to RF signals, which randomly masks some queries in $\mathbf{x}_{spatial}$ to activate the ability

of modeling non-local multi-head attention. The final input can be represented as:

$$\mathbf{X}_{spatial} = [\mathbf{x}_{spatial}^1; \mathbf{x}_{spatial}^2; \cdots; \mathbf{x}_{spatial}^D] \quad (20a)$$

$$\mathbf{x}_{spatial} = [\mathbf{x}^1; \cdots; \text{Mask}(\mathbf{x}^i); \cdots; \mathbf{x}^J] + \mathbf{PE}_{sp} \quad (20b)$$

where $\mathbf{PE}_{sp} \in \mathbb{R}^{J \times C}$ is the learnable position embedding for each joint query.

The architecture of SAM is depicted in Fig. 6. Since our task is to directly regress the 3D skeletons of multiple person instances, we apply a progressive dimension reduction scheme to our attention modules. Specifically, SAM contains N blocks in which each block contains a multi-head attention encoder for non-local spatial dependency modeling and a linear layer for dimension reduction. Skip-connection method is also applied in each block to make the training more stable. Following this scheme, the model can gradually regress the output of each module to the final 3D coordinates.

3) *Temporal Attention Module (TAM)*: Although SAM can reconstruct the complete 3D pose from a single RF snapshot, it may be inaccurate due to the lack of information in the time dimension. Hence, the Temporal Attention Module is proposed to refine the 3D skeleton sequences based on the temporal dependencies. Denote the output of SAM as $\mathbf{H}_{spatial}$. Similarly, we transform $\mathbf{H}_{spatial}$ to focus on the time dimension:

$$\mathbf{H}_{spatial} \in \mathbb{R}^{T \cdot M \times J \times C'} \Rightarrow \mathbf{X}_{temporal} \in \mathbb{R}^{M \times T \times \tilde{C}} \quad (21a)$$

$$\text{where } \tilde{C} = J \cdot C' \quad (21b)$$

in which each feature $\mathbf{x}_{temporal} \in \mathbb{R}^{T \times \tilde{C}}$ of $\mathbf{X}_{temporal}$ is the temporal queries of each person instance in the RF sequence.

The other change is the Temporal Mask. We mask the queries of random frames and make the module to predict all 3D skeleton sequences, which enforces the module to facilitate non-local interactions in the time dimension. The input can be written as:

$$\mathbf{X}_{temporal} = [\mathbf{x}_{temporal}^1; \mathbf{x}_{temporal}^2; \cdots; \mathbf{x}_{temporal}^M] \quad (22a)$$

$$\mathbf{x}_{temporal} = [\mathbf{x}^1; \cdots; \text{Mask}(\mathbf{x}^j); \cdots; \mathbf{x}^T] + \mathbf{PE}_{tp} \quad (22b)$$

where $\mathbf{PE}_{tp} \in \mathbb{R}^{T \times \tilde{C}}$ is the learnable position embedding for each frame query.

D. Training Loss

Since RPM 2.0 predicts a fixed size of M 3D skeletons in each RF image, where M is set to be larger than the actual number of people, the main challenge is how to associate the predictions correctly for different person instances. This is solved by a bipartite-matching-based training strategy.

1) *Bipartite Matching*: RPM 2.0 estimates the corresponding 3D skeletons based on the detection results. Therefore, we utilize the detected bounding boxes for matching.

Denote the predicted bounding boxes as $\{\mathbf{B}_m\}_{m=1}^M$ and the corresponding detection confidence scores as $\{p_m\}_{m=1}^M$ which derived from the center heatmaps. The ground truth set $\hat{\mathbf{B}}$ is padded with empty element $\emptyset$ to keep the same size M with the predicted set. Afterward, we find an optimal permutation

of $\sigma^* \in \aleph_M$ between the predicted set and the ground truth set that achieves the lowest matching cost:

$$\sigma^* = \arg \min_{\sigma \in \aleph_M} \sum_{m=1}^M \mathcal{L}_{\text{match}}(\hat{\mathbf{B}}_m, \mathbf{B}_{\sigma(m)}) \quad (23)$$

The matching cost can be computed by the bounding boxes and the detection confidence scores:

$$\begin{aligned} \mathcal{L}_{\text{match}}(\hat{\mathbf{B}}_m, \mathbf{B}_{\sigma(m)}) = & -p_m + L_{\text{dist}}(\hat{\mathbf{B}}_m, \mathbf{B}_{\sigma(m)}) \\ & + L_{\text{GIoU}}(\hat{\mathbf{B}}_m, \mathbf{B}_{\sigma(m)}) \end{aligned} \quad (24)$$

where $\hat{\mathbf{B}}_m \neq \emptyset$, L_{dist} and L_{GIoU} computes the $L1$ distance and Generalized IoU [63] between ground truth and predicted bounding boxes, respectively. Motivated by previous works [64], [65], we apply the Hungarian algorithm [66] to compute the optimal assignment of $\sigma^* \in \aleph_M$ with the matching cost.

2) *Loss Function*: We first compute the detection loss of RPM 2.0:

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{center}} + \mathcal{L}_{\text{offset}} + \mathcal{L}_{\text{size}} \quad (25)$$

Given the obtained optimal assignment σ^* , we can compute the 3D pose estimation loss:

$$\mathcal{L}_{\text{pose}} = \sum_{m=1}^M \{\hat{\mathbf{Y}}_{m \neq \emptyset}\} \mathcal{L}_{\text{pose}}(\hat{\mathbf{Y}}_m, \mathbf{Y}_{\sigma^*(m)}) \quad (26)$$

where $\hat{\mathbf{Y}}$ and $\mathbf{Y}$ are the ground truth 3D poses and predicted 3D poses, respectively. $\mathcal{L}_{\text{pose}}$ computes the average $L2$ distance between absolute predicted 3D skeletons and ground truth skeletons.

Our overall objective function is written as:

$$\mathcal{L} = \mathcal{L}_{\text{pose}} + \lambda \mathcal{L}_{\text{det}} \quad (27)$$

where λ balances the two loss terms, which is set to 0.1 in our experiments.

V. EXPERIMENTS

A. Implementation Details

Hardware: We use two TI AWR2243 commercial millimeter-wave radars to form our radar system, which has 12 transmitters and 16 receivers with a bandwidth of 1.23 GHz. The multi-camera system consists of 13 nodes, each of which is a Raspberry Pi 4 with a wide-angle lens that is powered by a Power-over-Ethernet (PoE) module and synchronized with the NTP server in the LAN. All of the cameras are calibrated with respect to the global coordinate system using standard camera calibration techniques [67]. The entire data acquisition system is controlled by a single laptop and can be assembled quickly in new scenarios.

Datasets and Metrics: To train and evaluate our RPM 2.0 framework and other baseline methods, we collect a large-scale multimodal dataset where the multi-view cameras provided the labels and the millimeter-wave radars provided the RF data. We ask 10 volunteers to walk randomly in 10 different scenarios when collecting data. To decouple the subjects and the environment, each volunteer is of different height, age, and

TABLE I
THE MPJPE METRIC (UNIT: *mm*) COMPARISONS AMONG STATE-OF-THE-ART METHODS AND RPM 2.0 FRAMEWORK. NOTE THE SINGLE-PERSON DATA IS THE SUBSET OF THE MULTI-PERSON DATA. BOLD FONTS REPRESENT THE BEST RESULTS.

Method	Nose	Neck	Sho	Elb	Wri	Hip	Knee	Ank	Mean ($\downarrow$)
Single-Person									
RFPose3D [50]	100.5	73.7	117.2	149.7	159.4	107.6	149.5	168.0	134.1
RFPose-OT [53]	86.9	69.8	84.7	108.5	111.0	84.6	110.4	122.4	100.0
RPM [52]	57.7	49.0	52.4	64.6	65.8	51.5	60.5	65.8	59.2
RPM 2.0, concat	55.8	37.0	50.5	69.1	71.8	47.4	60.0	66.1	58.4
RPM 2.0, sum	56.1	37.9	52.6	70.3	74.5	48.4	59.9	67.2	59.8
RPM 2.0, softsum (Ours)	55.4	37.0	49.8	67.2	69.4	47.1	58.3	64.8	57.5
Multi-Person									
RFPose3D [50]	114.0	84.3	133.0	162.7	175.5	117.7	155.7	172.5	145.2
RPM 2.0, concat	71.9	53.3	65.4	83.7	86.3	62.0	74.0	81.6	74.6
RPM 2.0, sum	73.1	53.3	69.6	88.5	90.5	63.8	76.6	83.4	76.5
RPM 2.0, softsum (Ours)	69.2	50.3	65.7	84.8	85.7	61.0	73.5	80.4	73.0

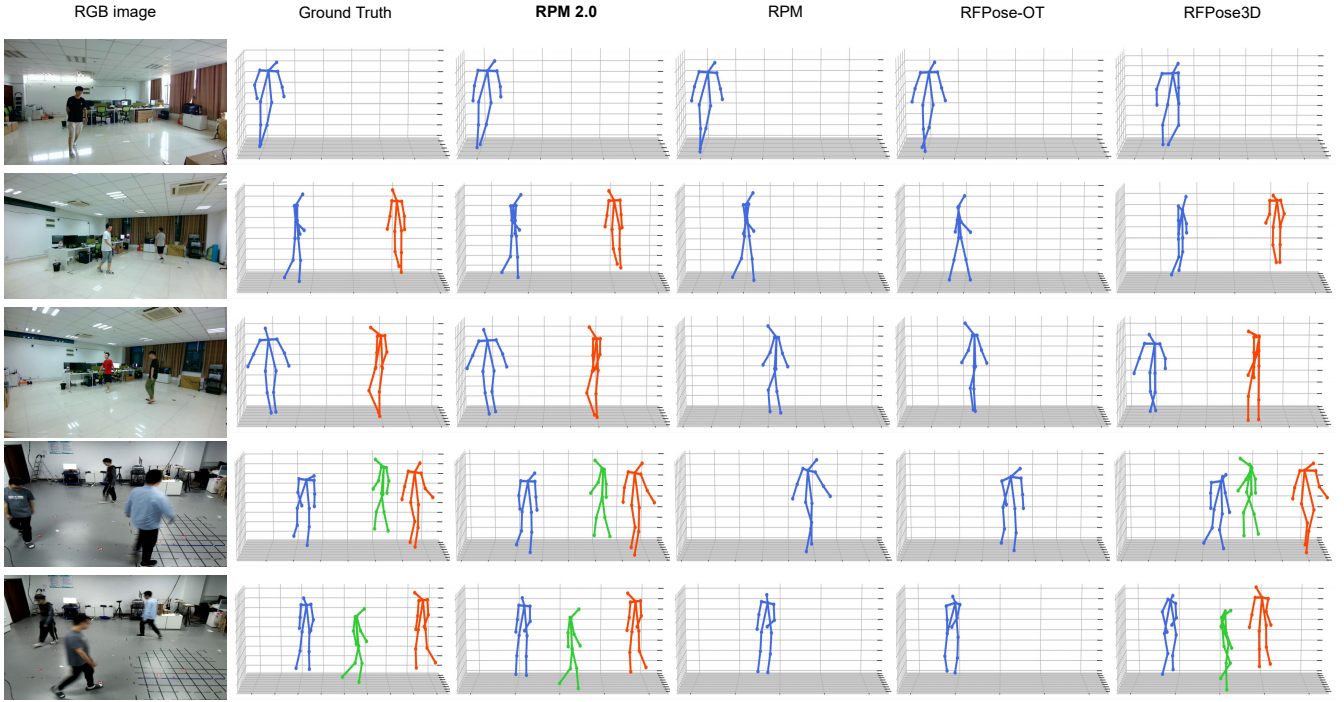


Fig. 7. Qualitative results under basic environments. From left to right are the images taken by the camera, the 3D pose labels, the poses generated by RPM 2.0, RPM, RFPose-OT and RFPose3D, respectively.

dress and is randomly grouped thus avoiding network learning to correlate subject and environment. The final dataset consists of 516,840 RF images and the corresponding annotations, of which 443,680 RF images are used to train the network and the remaining 73,160 RF images are used to evaluate the network performance.

We use the absolute Mean Per Joint Position Error (MPJPE) [68] to evaluate the performance of RPM 2.0 and other baseline methods, which measures the L_2 distances between the ground truth joints and the predicted joints in the world coordinate.

Training: We implement RFPose3D [50], RPM [52] and RFPose-OT [53] as the baseline methods. Since RPM and RFPose-OT can only estimate 3D skeleton of single person, we train and test these two methods on a subset of the collected dataset in which only one single person walks randomly. All

methods are trained on 3 NVIDIA A100 GPUs. We train RPM 2.0 for 200 epochs, with a batch size of 32 and a learning rate of 3×10^{-4} . The cosine annealing with a warmup of 40 epochs is adopted as the learning rate scheduler.

B. Experimental Results

1) *Performance under basic environments:* The quantitative results are shown in Table I. Several findings can be derived from the table. First of all, All methods have improved performance on the single-person data set compared to the multi-person data. This may be due to the fact that multi-person datasets are more complex and variable, thus making pose estimation more difficult. RPM [52] and our proposed RPM 2.0 have both performed well due to the novel network design. Secondly, due to the fast movement and small reflex area of limbs like wrists and ankles, all methods showed a

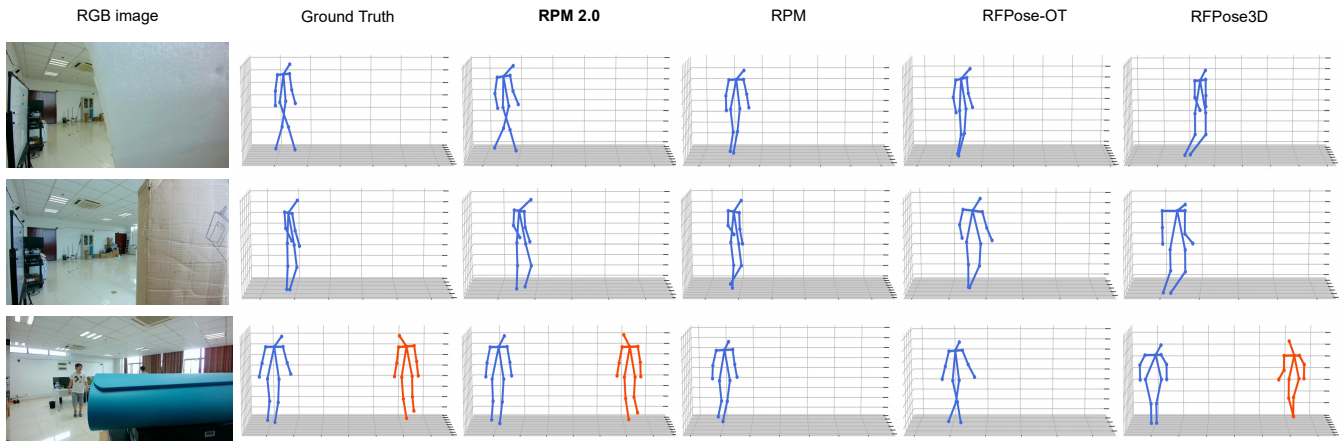


Fig. 8. Qualitative results of RPM 2.0 and other baseline methods under occlusion environments. From top to down are environments where radars are occluded by a form, a carton and a yoga mat, respectively.

decrease in accuracy in predicting these body parts. Moreover, different feature aggregation methods have different effects on the performance of RPM 2.0. The **softsum** method, which sums multi-view features based on confidence achieves better performance than the other two methods. We speculate that this might be due to the fact that confidence-based weighted fusion discards the wrong feature from a single view, which leads to a more robust feature representation. Finally, the worst performance of RFPose3D [50] may be due to the different data distribution. Specifically, RFPose3D utilized a custom FMCW radar with a large antenna array. By contrast, we utilize portable and commercial millimeter radars for data collection, which are vulnerable to various kinds of interference in practical environments. Hence, it is more difficult to estimate accurate human pose from our collected RF data.

To better understand the difference in performance, we provide a visual comparison between RPM 2.0 and other baseline methods. As illustrated in Fig. 7, RPM 2.0 is able to reconstruct accurate multi-person 3D skeletons. RPM and RFPose-OT perform well on single-person data, but struggle with multi-person data due to model limitations. RFPose3D often produces some abnormal poses and performs not very well in locating the person instances. Besides, it can be observed that the data of three-person walking is a greater challenge for RPM [52] and RFPose-OT [53]. Since these two methods can only predict the pose of a single person, the presence of other bodies in the scene can interfere and result in a significant deviation in the position of the predicted 3D pose in space. In addition, RFPose-OT [53] shows a large degradation in pose accuracy. Although RFPose3D [50] can predict multi-person poses, its performance on three-person walking data is not good enough, resulting in significant distortion of the predicted pose. On the other hand, our proposed RPM 2.0 demonstrates superior performance on these challenging data due to its well-designed network architecture, accurately localizing the human body in space and predicting the 3D pose.

2) *Performance under occlusion or bad lighting condition:* We also show the qualitative results of our proposed framework under occlusion or low illumination environments, where

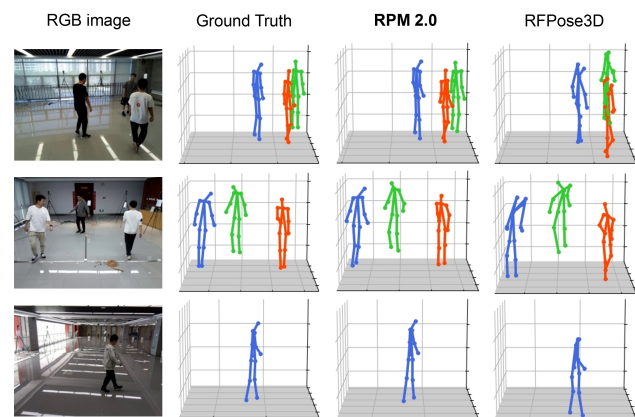


Fig. 9. Qualitative results of RPM 2.0 and RFPose3D [50] under new scenarios.

the camera-based solutions usually have poor performance. Fig. 8 and Fig. 1 show some illustrative examples. As we can see, in occlusion scenarios, the constructed skeletons of RPM 2.0 are almost the same as the ground truth, which demonstrates the advantage of our proposed framework over the vision-based solutions and other baselines.

TABLE II
THE MPJPE METRIC (UNIT: *mm*) COMPARISONS ON THE NEW DATA AMONG STATE-OF-THE-ART METHODS AND OUR FRAMEWORK. BOLD FONTS REPRESENT THE BEST RESULTS.

Method	Mean ($\downarrow$)
RFPose3D [50]	253.4
RPM 2.0 (ours)	116.2

3) *Generalization to New Scenarios:* In order to evaluate the cross-domain generalization of RPM 2.0, we collected more data in 3 new outdoor scenarios and directly apply the trained models to new scenario and evaluate the performance of all methods. As shown in TABLE II, all methods show a performance degradation for these new data. However, RPM 2.0 still achieves an average error of 116.2 mm, which

confirms the generalization capability of our model.

We also visualize some qualitative results on the new collected data to better understand the performance of RPM 2.0. As shown in Fig. 9, RPM 2.0 accurately estimates the 3D poses of the new subjects, which shows good generalization performance. RFPose3D [50] shows larger errors when dealing with data in new scenes, and sometimes fails to generate reasonable skeletons.

TABLE III
ANALYSIS ON THE MODEL ARCHITECTURE.

Backbone	MFN	SAM	TAM	MPJPE ($\downarrow$)
HRNet	×	×	×	174.1
	×	✓	×	127.5
	×	×	✓	120.2
	×	✓	✓	95.7
	✓	×	×	155.0
	✓	✓	×	102.4
	✓	×	✓	98.6
	✓	✓	✓	73.0

TABLE IV
ANALYSIS ON DIFFERENT SIGNAL INPUT.

Method	Hor	Ver	MPJPE ($\downarrow$)
RPM 2.0	✓	×	103.2
	×	✓	132.4
	✓	✓	73.0

C. Ablation Study

1) *The Design of RPM 2.0*: We conduct extensive ablation studies to evaluate the contribution of individual components of RPM 2.0. As shown in Table III, after removing the MFN module, the network's MPJPE metrics all show some degree of degradation. Specifically, the performance of the model with SAM and TAM but without MFN is 31% lower than the full model, achieving 95.7mm, which shows the effect of MFN. When either SAM or TAM is applied, the performance of the network is substantially improved. Interestingly, we observe that the network using only TAM performs slightly better than the one using only SAM. This is probably because TAM can use temporal information to smooth the pose of each frame.

We also explored the impact of inputs from different perspectives on network performance. As shown in Table IV, RPM 2.0 has achieved competitive performance when using only the RF signal generated by the horizontal radar as input. However, the performance is a little worse when only vertical RF signals are utilized. When both views of the RF signal are used, RPM 2.0 obtains the best performance, achieving 73.0mm.

TABLE V
DETECTOR SIZE AND COMPUTATION COMPARISON.

Methods	Detection time (ms)	AP	MPJPE (mm)
RPN [69]	3.2	0.72	74.6
DETR [65] (2 stage)	15.6	0.76	75.2
anchor-free detector (ours)	0.4	0.75	73.0

The detector is also important for the final performance because the area occupied by the human body in RF signals is usually small, and if the target cannot be detected from the sparse RF signals, it is difficult to predict the human pose. Also, the accuracy of the detection directly affects the final loss calculation since the final loss function calculation requires matching the labels and predicted poses based on the detection results. Therefore, we also explore the impact of different detection modules. Specifically, we replace the proposed anchor-free detector with the RPN module in Faster RCNN [69] or DETR [65] and evaluate their performance. Note that since the number of DETR parameters is too large leading to very unstable end-to-end training, we first train DETR on RF data and then integrate it into the RPM 2.0 framework. As shown in Table V, RPN module performs well in detecting the human position, achieving an average precision (AP) of 0.72. Although DETR performs best in human detection, it does not perform well in pose estimation because it cannot be trained end-to-end and the backbone network cannot generate robust feature representations for subsequent pose estimation. Our proposed anchor-free detector, since it does not require a large number of anchors like RPN, has the fastest detection speed and ensures good detection accuracy, achieving an AP of 0.75. Furthermore, it can be trained end-to-end with the subsequent modules and therefore achieves the best pose estimation performance.

TABLE VI
ANALYSIS ON HYPERPARAMETERS.

Input Length	10	20	30	40	50
MPJPE ($\downarrow$)	79.7	73.0	73.4	75.5	74.9
Joint Mask Percentage	10%	20%	30%	40%	50%
MPJPE ($\downarrow$)	75.0	75.1	73.0	73.7	79.5
Frame Mask Percentage	0%	10%	20%	30%	40%
MPJPE ($\downarrow$)	73.4	73.0	75.7	75.5	81.0

2) *Architecture Parameter Analysis*: To achieve the best performance, we explore the various hyperparameter settings. We first report the ablation experiment results on the input RF sequence length. As illustrated in Table VI, we observe that a sequence length of 20 frames is optimal. Too long or too short sequences can cause performance degradation.

To better model non-local relationships among spatial joints and frames, we mask some joint queries or frame queries in the training process. We also report the effect of different mask ratios. From Table VI, we can observe that RPM 2.0 achieves the best performance when 30% joint queries are randomly masked. Nevertheless, exceeding this ratio can result in excessive spatial information loss, which leads to performance degradation. Similarly, the frame mask percentage of 10% is optimal for RPM 2.0.

D. Inference Speed Comparison

We also report the comparison results about the computational complexity and inference speed. In this study, two common complexity metrics, model parameters (Params) and multiply-accumulate operations (MACs), are utilized to quantify the complexity of the different models examined. In

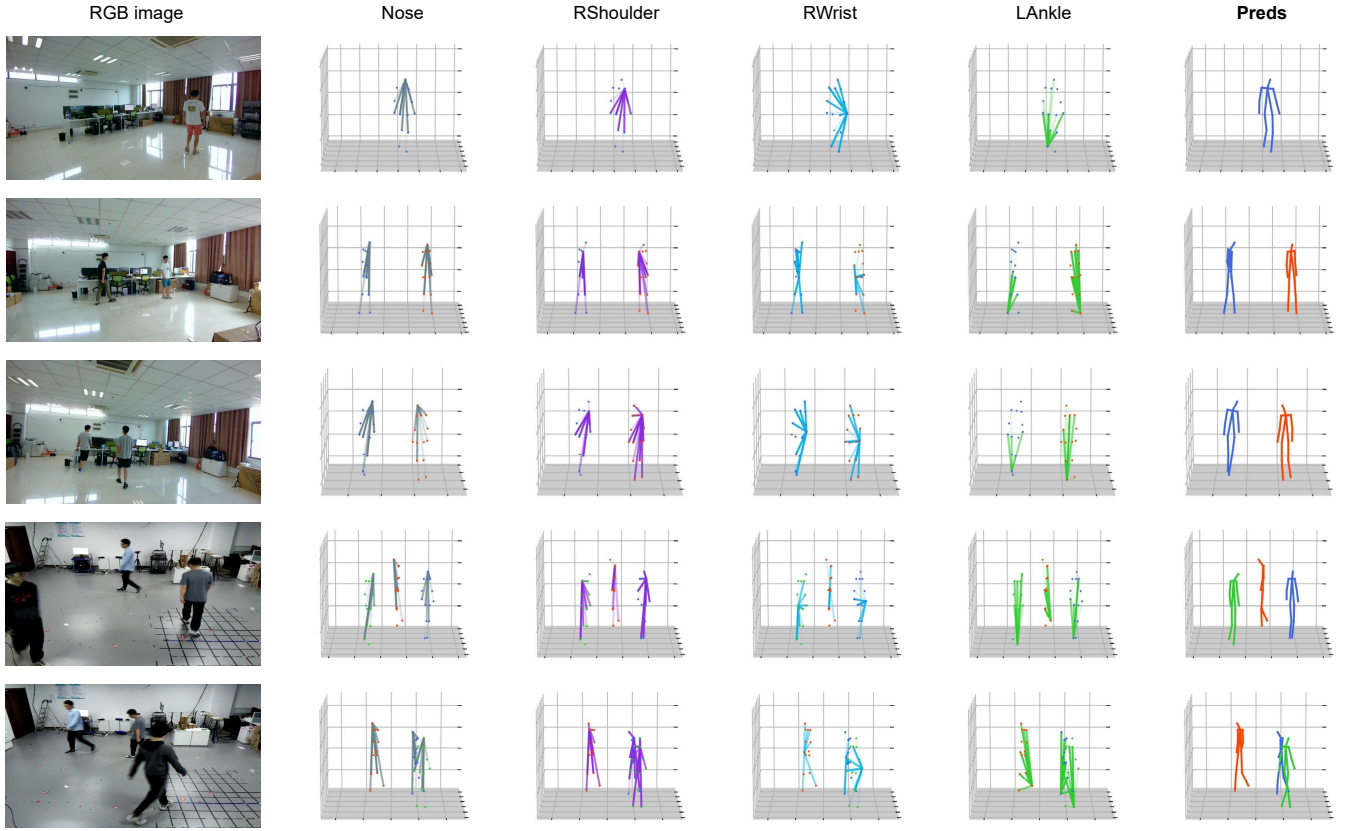


Fig. 10. Visualization of spatial attention in RPM 2.0. We show the self-attentions between specified joints and all other body joints, where darker color indicates stronger attention.

TABLE VII
MODEL SIZE AND COMPUTATION COMPARISON.

Methods	Params (M)	MACs (G)	Speed (ms)
RFPose-OT [53]	7.02	1.30	29
RFPose3D [50]	10.92	25.33	32
RPM [52]	81.67	1375	49
RPM 2.0 (Ours)	38.41	184.93	43

addition, the inference speed of each method is calculated to reflect their computational efficiency. As shown in Table VII, RFPose-OT [53] is found to be the most lightweight model in terms of both model size and complexity, with the fastest inference speed. RFPose3D [50] shows superior performance in terms of model size and inference speed, as it was modified based on the ResNet18 architecture. However, these models exhibited only moderate accuracy in pose estimation. RPM [52] uses spatio-temporal attention networks for single-person pose estimation and applies self-attention operations to the complete RF signal map. Despite the adoption of dimensionality reduction, the model complexity of RPM remains very high, but its inference speed is still relatively fast, taking only 49ms per batch of data. The proposed RPM 2.0 also uses spatio-temporal attention mechanisms. However, benefited from more efficient and lightweight network designs and a processing flow that involved detection before estimation, RPM 2.0 only needs to crop out the region of interest features without processing the entire signal map, resulting in

a clear advantage in terms of model parameters, computational complexity, and inference speed compared to RPM. Therefore, RPM 2.0 achieves a good balance between speed and accuracy among all the methods studied.

E. Attention Visualization

In order to better understand the effect of RPM 2.0 in modeling spatio-temporal attention, we also visualize the spatial and temporal interactions in our framework.

Fig. 10 illustrates the visualization of spatial attention. We show the self-attentions between specified joints and all other body joints. The darker color indicates stronger attention. We observe that RPM 2.0 attends to adjacent body parts when reconstructing specific joints. In the first row, when predicting the location of the right shoulder, RPM 2.0 pays more attention to relevant non-local joints, especially those on the left arm and right wrist, which provide strong cues to the shoulder location. Besides, RPM 2.0 adjusts self-attention based on different inputs and person instances. For example, in the second row, RPM 2.0 only focuses on a few relevant body parts when estimating the right wrist location of the right person, which is different from the attention in the other rows and person instances.

Furthermore, we also show the temporal attention in RPM 2.0. As indicated in Fig. 11, we visualize the temporal attention between a specified moment of T (the middle column) and other frames. Similarly, we observe that RPM 2.0 adaptively

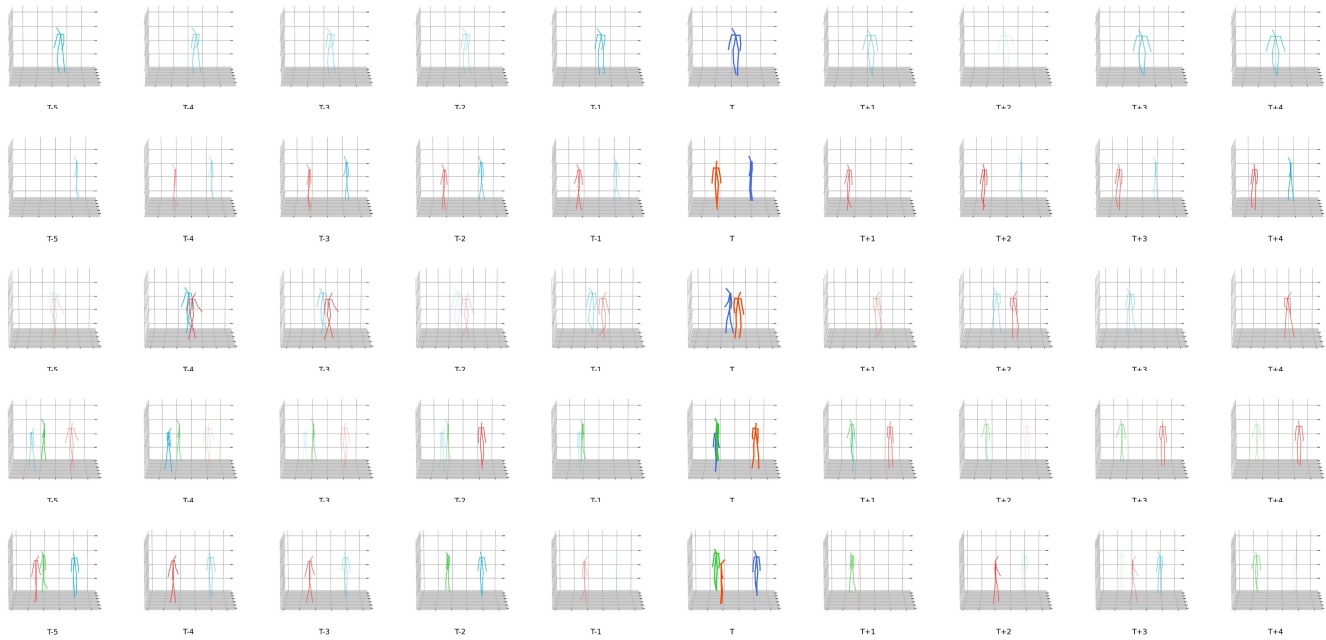


Fig. 11. Visualization of temporal attention in RPM 2.0. Note that we visualize the temporal attention between a specified moment of T (the middle column) and other frames, where darker color indicates stronger attention.

models temporal dependencies based on specific sequences. In addition, we can observe from the visualization results that RPM 2.0 does not have the problem of long memory loss. Thanks to the use of self-attention mechanism to model temporal correlation, RPM 2.0 can focus on all the information in the sequence simultaneously without considering their distances in the sequence.

human body in space, and the predicted pose was distorted due to the degradation of the signal quality.

2) *Rare poses*: So far, due to the limitation of the dataset, RPM 2.0 can only estimate the posture of walking people. RPM 2.0 does not reconstruct the accurate pose well when the subject has rare poses (Fig. 12 (c)). More diversity in the dataset and hard-example mining could improve the network performance.

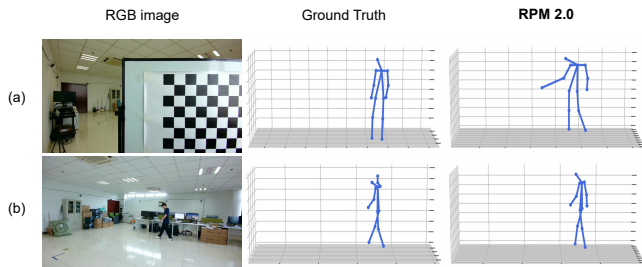


Fig. 12. Examples of failure cases: (a) Interference from strong reflectors; (b) Rare poses.

F. Failure Cases

We observe several failure cases in our experiments (see Fig. 12).

1) *Interference from strong reflectors*: Although RF signals can traverse many occlusions, strong reflectors like metal can cause serious interference, resulting in a dramatic degradation of signal quality. As shown in Fig. 12 (a), when the metal whiteboard is blocked in front of the radar, the network prediction deviates significantly from the ground truth. In particular, RPM 2.0 could not predict the real position of the

VI. CONCLUSION

In this paper, we proposed RPM 2.0, a novel framework for multi-person 3D pose estimation using RF signals. The Feature Extraction Network located person instances via a lightweight anchor-free detector and extracted high-resolution features. The Multi-view Fusion Network fused features from horizontal and vertical mmWave radars in a unified latent space. Finally, the Spatio-Temporal Attention Network reconstructed accurate 3D skeleton sequences based on non-local spatio-temporal relationships. The experimental results demonstrated that RPM 2.0 was able to estimate multi-person 3D poses with an average error of $73mm$ and maintained its performance when generalized to new data such as occlusion or low illumination.

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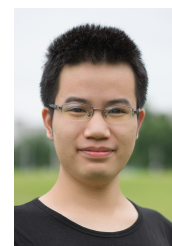
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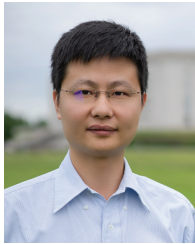
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